Workshop Pole Barn: Building the Posts

How to nail up the 3-ply 2×8 posts on site · 19 posts, cut from 16 ft boards · Prairie, Idaho · planned for 2028

> **This is the fastening schedule, not a stamped design.** It follows the American Wood Council's National Design Specification for Wood Construction (NDS), Section 15.3, so the three boards can be counted as one built-up column. It does not prove that a post of this length at 8 ft spacing can carry this building's 120 psf snow and 115 mph wind. An Idaho engineer still has to accept the post, the bracket, and the bolts. Take this page to that review.

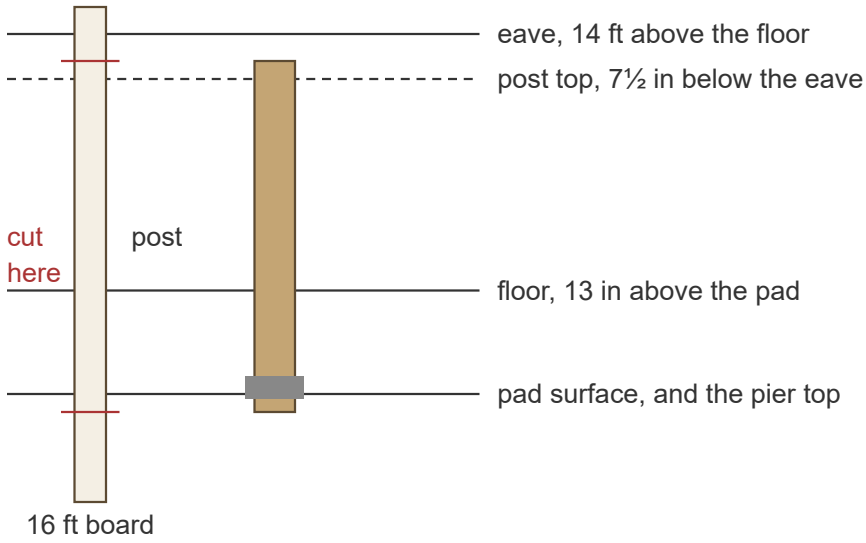
What you are building

Each post is three 2×8s nailed face to face, about 4½ in by 7¼ in, standing on a wet-set bracket just above the pad and stopping 7½ in short of the truss beam at the 14 ft eave. The plans turn the post so the **7¼ in face goes through the wall** and the 4½ in face is the one the girts land on. Nail into the 7¼ in face. Do not nail into the 4½ in edge.

Why the boards are 16 ft

The 14 ft figure is the wall, from the floor up to the eave. It is not the post. The post starts on the bracket, down at the pad, and the floor sits 13 in above that pad. The post then stops 7½ in short of the eave, under the truss heel. From the pad surface up to that top is 14 ft + 13 in − 7½ in = **14 ft 5½ in**. A 14 ft board is 5½ in short of that, before the bracket lifts the wood off the concrete. Yards sell 2×8s in even lengths, so the order is 16 ft. Cut the extra off. Do not stand a 16 ft post in a 14 ft wall.

Figure 1. The 16 ft board is stock. The post is the part from the bracket seat up to 7½ in below the eave. Cut to that length before nailing, so the end nails land 3½ in from the real ends.



Measure one bracket before you cut the pile. The seat is a little above the pad, so the board is a little shorter than 14 ft 5½ in. Cut all three plies to that measured length, ends square and flush. Then nail. If you nail the 16 ft board and saw the end off afterward, the end nails will not sit 3½ in from the cut, and that end of the post is not on this schedule.

- **Boards.** Buy 2×8 × 16, No. 2 or better SPF, then cut them to the measured post length. Three plies per post, each one the full post, no splices: NDS 15.3.1 counts a built-up column only when every ply runs the whole post. Buy about 60 boards for the 19 posts (57, plus a few to cull). Reject twist, crook, and any set of three that would stack a big knot in the same spot.
- **Treated or not.** The materials list's site-build note wanted the bottom 4 ft pressure-treated. A treated stub spliced onto untreated boards is a splice, so it is not this schedule. If the yard has straight treated 2×8×16 in No. 2, use those for all three plies. Treated lumber is often wet and crooked: sticker it flat, with weight on the pile, until it stops dripping, then build. If 16 ft treated stock is not straight, use untreated No. 2 SPF for the full length. The post sits on the bracket above the pad, and the 2×10 pressure-treated splash board is the treated piece at grade.
- **Glue.** PL Premium on the two faces that touch, then clamp until the faces are in full contact. The glue holds the boards while you nail and fills small gaps. It does not replace the nails, and it does not make this a glulam.

The nail

Use a **30d common wire nail, 4½ in long, 0.207 in diameter**, hot-dipped galvanized. That is the nail NDS Figure 15C uses for a 3-ply column. A 3-ply post is 4½ in thick, and the nail has to pass through two plies and at least three-quarters of the third: 2 × 1½ in + ¾ × 1½ in = 4⅛ in. A 30d nail is 4½ in, so the tip reaches the far face and should not stick out. A 16d nail is 3½ in and stops inside the third board. Do not use it.

- **Not a gun nail.** A 30d common nail does not fit an ordinary framing nailer. Drive it by hand. If the wood splits, mostly near the ends and in treated lumber, predrill with a 3/16 in bit. That is smaller than the shank, so the nail still bites. Do not drill a hole as big as the nail.
- **Galvanized.** Hot-dipped, not electro-galvanized. The post bottom sits in the splash zone, and a bright nail rusts there. Electro-galvanized is a thin coat.
- **Not screws, unless the screw has its own report.** Deck screws, drywall screws, and ordinary construction screws are the wrong fastener: they snap in shear. Simpson's published screw schedule for built-up columns (Strong-Drive SDW Truss-Ply) is drawn for 2×4 and 2×6 posts, not for a 2×8, so it is not copied here. A structural screw may replace these nails only on the schedule in that screw's own evaluation report, for this exact 3-ply 2×8.

The pattern on a 2×8

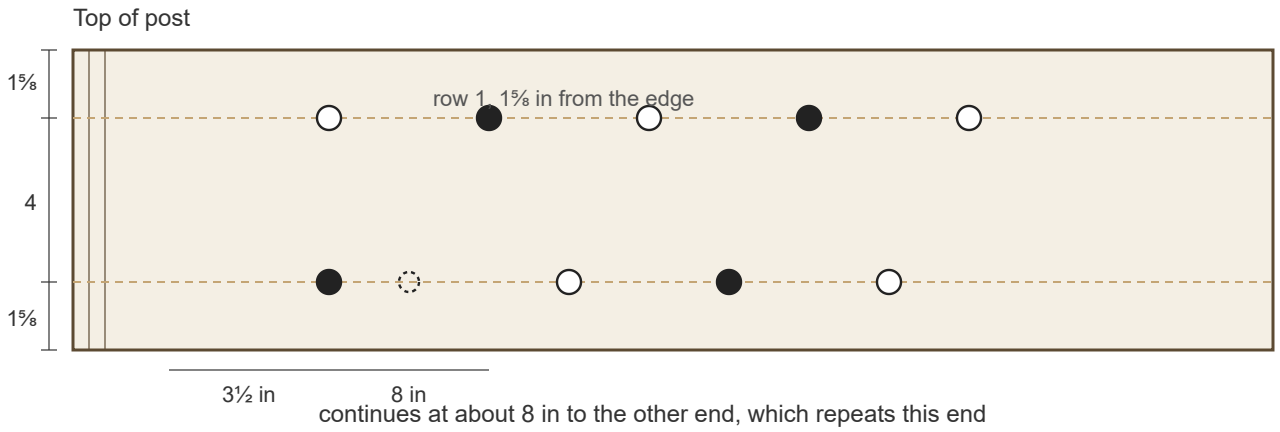
NDS Figure 15C draws the typical schedule for a 3-ply 2×4 (one row) and a 3-ply 2×6 (two rows). It does not draw a 2×8. The 2×6 drawing cannot be copied onto this post: its rows sit 1½ in from each edge, and on a 7¼ in face those two rows would be 4¼ in apart. The rule limits the rows to 20 nail diameters, which is 4.14 in for this nail. The pattern below is the same rule set, fitted to the 7¼ in face.

Where the numbers come from (30d common nail, 0.207 in diameter; ply thickness 1½ in)

Rule (NDS 15.3.3)	The limit	What this page uses
How many rows	One row when the face is no wider than 3 × 1½ in = 4½ in (the 2×4 in Figure 15C). Two rows when it is wider (the 2×6). A 2×8 face is 7¼ in, so two rows, not three.	Two rows
Distance from the edge	5 to 20 diameters: 1.04 in to 4.14 in	1⅝ in from each edge
Distance between the rows	10 to 20 diameters: 2.07 in to 4.14 in	4 in (1⅝ + 4 + 1⅝ = 7¼)
Distance along a row	20 diameters to 6 × 1½ in: 4.14 in to 9 in. Figure 15C uses 8 in.	8 in, or a hair over, so the end nails still land
Distance from each end	15 to 18 diameters: 3.1 in to 3.7 in	3½ in
Which face	Adjacent nails are driven from opposite sides, and each nail passes through all three plies.	Alternate faces along each row

Mark the pattern on the cut post, not on the 16 ft board. On a 14 ft 5½ in post the end nails are 3½ in from each end, 166½ in apart. Twenty-one equal spaces in that 166½ in are 7.9 in, inside the 4.14 in to 9 in limit, so each row gets 22 nails. A post cut a little shorter, because the bracket seat is above the pad, gets the same end nails and the same spacing, and one fewer nail in a row if the last space would pass 9 in. Mark both rows from the same end. Put the second row's field nails halfway between the first row's, about 4 in along, but keep a nail in each row 3½ in from each end. At those shared end spots, drive the two nails from opposite faces.

Figure 2. The 7¼ in face of the cut post, nails at the ends and the first few of each row. Open circles are driven from the near face; filled circles from the far face. The rows then continue at about 8 in to the other end.



The dashed circle is the 4 in offset, not an extra nail. Open and filled circles are opposite faces. Both rows still have a nail 3½ in from the end.

How to build one post

1. Pick three straight boards and cut them to the measured post length (Figure 1) before any glue or nails. Sight along each one. Crown them the same way, crowns up, so they do not fight each other.
2. Work on a flat bench or on sleepers, long enough that the post cannot sag or twist while you nail. Spread PL Premium on the two inner faces and clamp the three boards until the faces touch along the whole length. NDS 15.3.1 wants the faces in contact.

3. On the 7¼ in face, snap two lines, 1⅝ in from each edge. They will be 4 in apart. Mark 3½ in from each end on both lines, then divide the 185 in between those marks into 23 equal spaces (about 8 in). Offset the second line's field marks by half a space.
4. Drive a 30d hot-dipped galvanized common nail at every mark, alternating faces along each line. The head should finish flush. The tip should reach the far face and not blow out. If a tip does come through, clinch it across the grain.
5. Check the finished post. It should be 4½ in by 7¼ in, straight, and tight between the plies. Mark the 7¼ in face that goes through the wall, and which edge is the outside (the girt face), so it cannot be turned when it goes up.
6. Stand it on the bracket only after the bracket is set in the pier (footing page). The bracket has to be the one sized for a 3-ply 2×8. Do not plane the post to fit a different bracket, and do not set the wood in the concrete.

How many to buy

Item	Count	Notes
2×8×16, No. 2 SPF	60	57 boards for 19 posts, plus culls. This is the stock length. Cut to the measured post before nailing. No splices.
30d hot-dipped galvanized common nails, 4½ in	1,000	22 nails × 2 rows = 44 on a 14 ft 5½ in post, about 840 for 19 posts. A shorter cut drops a nail or two per row. The extra covers bent nails. About 50 lb. Not a gun nail.
PL Premium	12 tubes	The count already on the materials list. Two glue lines per post.
Wet-set bracket, 3-ply 2×8 size	19	Perma-Column or Sturdi-Wall Plus, as the materials list says. Set in the wet pier pour.

Sources

- American Wood Council, National Design Specification for Wood Construction, Section 15.3 (built-up columns) and Figure 15C (typical nailing schedules for 2-ply and 3-ply columns). Figure 15C specifies the 30d common nail, 4½ in long and 0.207 in in diameter, and draws two rows for a 3-ply 2×6. It does not draw a 2×8. The edge and row distances on this page are that section's spacing rules applied to a 7¼ in face, read November 2026. The secondary write-ups that check the reading are Simpson Strong-Tie's Designing Built-Up Columns (September 3, 2015) and the worked example at stepbystep.com.
- This building's post size, length, count, and orientation are the workshop model: 3-ply 2×8, the 7¼ in face through the wall, wet-set bracket, 16 ft, 19 posts.